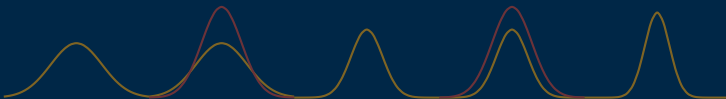


Gated Data-Space Inversion for Reservoir Prediction

Reducing variance loss in posteriors

Masters dissertation ▷ Thiago Tomás de Paula



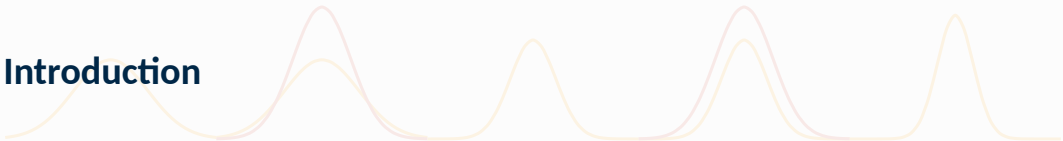
August 4th, 2026

Graduate Program in Mechanical Engineering

University of Brasília



Introduction



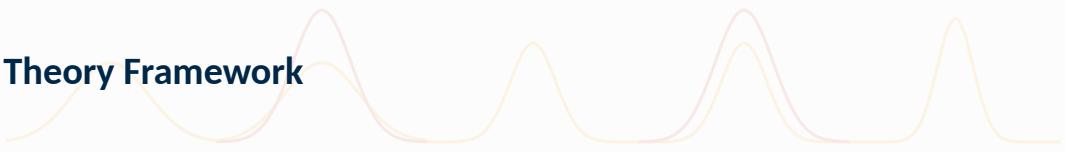








Theory Framework

The image features two decorative, wavy lines that span the width of the page. One line is a light yellow color, and the other is a light pink color. They are layered on top of each other, creating a soft, flowing background element. The lines have multiple peaks and valleys, giving them a rhythmic, wave-like appearance.



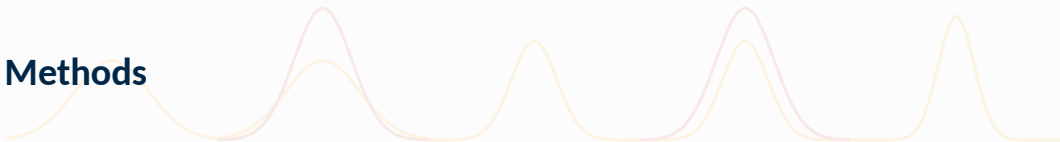








Methods



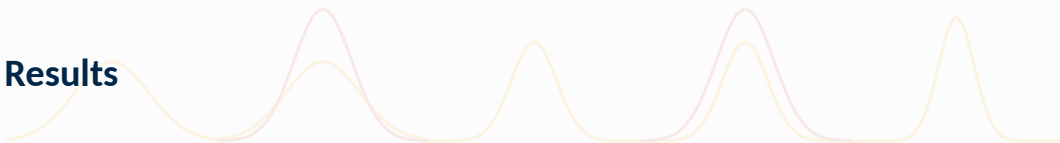


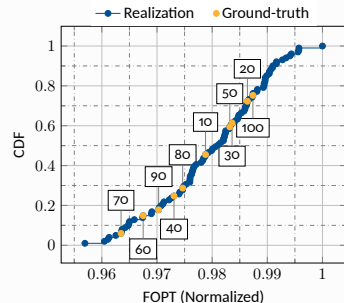
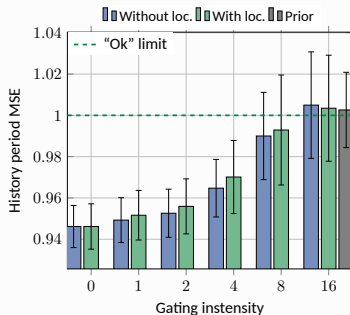
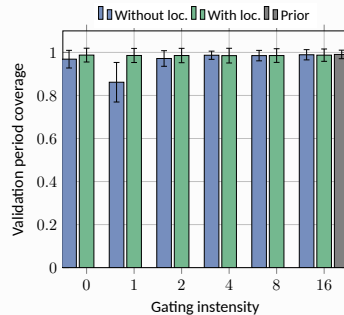
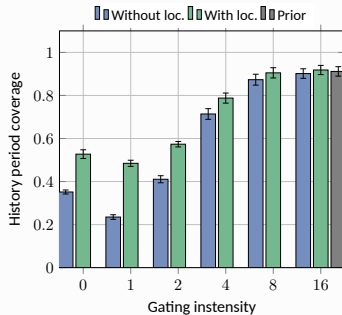


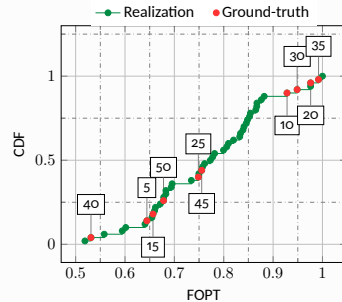
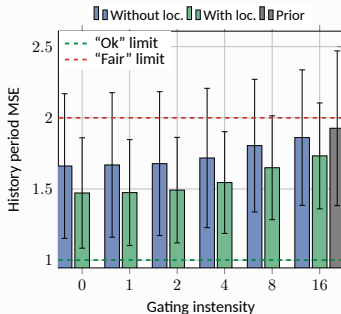
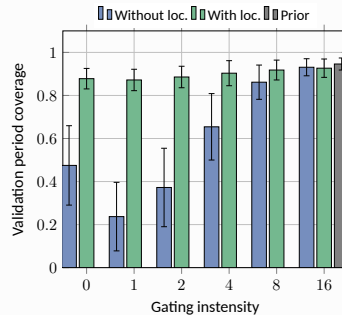
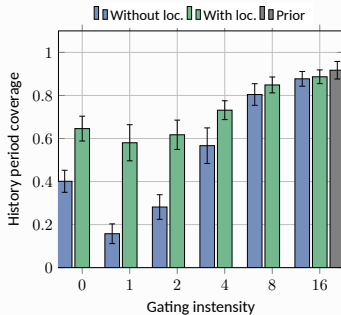


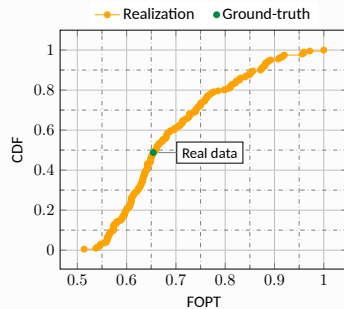
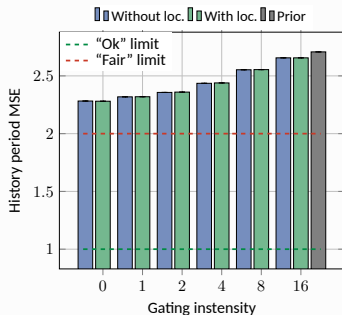
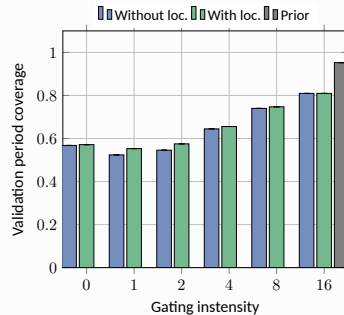
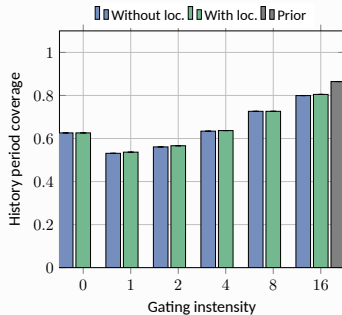


Results

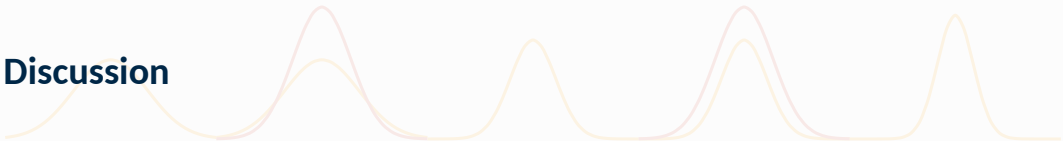








Discussion

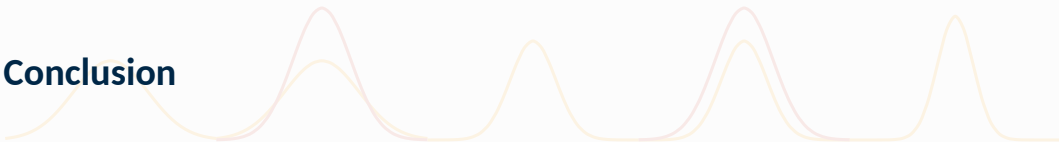








Conclusion











Gated DSI-ESMDA

Reducing variance loss in posteriors

Thiago Tomás de Paula

Graduate students at University of Brasília, Brazil

August 4th, 2026

“It’s hard to make predictions, specially about the future”

— Yogi Berra

Thank you!